USING IMPUTATION TECHNIQUES TO EVALUATE STOPPING RULES IN ADAPTIVE SURVEY DESIGN

by

Thais Piva
Duke University

Jerry Reiter
Duke University

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Abstract

Adaptive Design methods for social surveys utilize the information from the data as it is collected to make decisions about the sampling design. In some cases, the decision is either to continue or stop the data collection. We evaluate this decision by proposing measures to compare the collected data with follow-up samples. The options are assessed by imputation of the nonrespondents under different missingness scenarios, including Missing Not at Random. The variation in the utility measures is compared to the cost induced by the follow-up sample sizes. We apply the proposed method to the 2007 U.S. Census of Manufacturers.

Stopping rules for adaptive survey designs using imputation techniques of multivariate continuous data with nonignorable missingness

Thais Paiva[1] and Jerry Reiter[2]

[1]Department of Computer Science, Federal University of Minas Gerais, Belo Horizonte, Brazil
[2]Department of Statistical Science, Duke University, Durham, NC, USA

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1 Introduction

Adaptive designs use auxiliary information to tailor and update the sampling scheme throughout a survey. This information may include administrative records; paradata, that is data about collecting the responses, such as details about where and how the interviews took place; and the actual survey data as they are collected. With these extra information available, the agencies conducting surveys can change strategies of collecting the data to improve the sample representativeness while allocating resources more efficiently (Miller, 2013; Finamore et al., 2013). Some of these changes affect the strategies applied to individuals, e.g., using different contact means or targeting underrepresented groups, but they can also apply to the entire sample, such as stopping the data collection. This later application is the subject of this paper.

The concept of adaptive designs originated in clinical trial studies, where different treatments are considered depending on characteristics of the patient and responses to previous treatments (Murphy, 2003) to increase response rates and avoid nonresponse bias. Some of these methods that consider different strategies for each individual were extended and applied to survey methodology in order to increase response rates and sample representativeness, e.g. R-indicators and partial R-indicators (Wagner, 2008; Schouten et al., 2009, 2011, 2012).

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Data collected at initial phases also can be used to guide decisions applied to general features of the survey design, not only to individually tailored strategies. For example, the agency can decide to stop data collection depending on the amount of information already recorded and the costs of completing data collection. The benefits of stopping data collection include reducing the total cost and releasing results earlier, but it only makes sense if the data quality and inference are not sacrificed to undesirable levels. The quality of data can be assessed by information measures based on quantities of interest and their estimated changes at each survey phase. Some stopping rules for surveys with multiple waves are proposed by Rao et al. (2008) and Wagner and Raghunathan (2010) for binary response variables. In both methods, the nonrespondents are imputed under the assumption that they follow the same distribution as the respondents; that is, the data are missing at random (MAR) (Rubin, 1976). In many sample surveys and observational studies, the ignorability assumption is invalid, so that standard methods for multiple imputation can produce unreliable estimates. For that reason, it is useful and important to develop methods for missing not at random (MNAR) data. As examples, Greenlees et al. (1982) proposes an imputation method for nonignorable response mechanism by considering a regression model with censoring, and Diggle and Kenward (1994) present a modeling approach for longitudinal data sets with a nonignorable dropout process.

We propose a method for handling nonignorable missingness when making multiple imputation of multivariate continuous data. To capture distributional features that this type of data may have, we use a mixture of multivariate normal distributions and a truncated Dirichlet process prior distribution. We fit the model to the set of observed data, resulting in an estimate of the respondents' distribution. When the data are missing not at random, we need a different distribution for the nonrespondents, following a pattern mixture model approach. To propose distributions for the nonrespondents, we alter the probabilities of the mixture components, keeping fixed the location and scale parameters. We inflate cluster probabilities to generate more points from that region, and deflate them to do the opposite, yielding a new distribution for the imputation. We then perform sensitivity analysis by examining different sets of altered probabilities. Since that is the main step of the imputation method, we developed an R application to implement the process of selecting new probabilities. The user selects new probabilities by setting the values on "sliders" for each components. The application automatically generates data from the specified distribution, and shows plots and summary statistics to be used for the sensitivity analysis.

Consider an ongoing survey that is being evaluated at a certain point in time to decide whether it is worth continue collecting data, or to stop it and impute the nonrespondents from what has been observed so far. This decision depends on the quantity and quality of the data that were already recorded, how different the nonrespondents are from the respondents, and how any differences could impact the inference results. To facilitate such decision-making, we propose some utility measures to estimate the change in the results under different missingness scenarios. Each scenario reflects one possibility for the missingness pattern, ranging from the MAR case, where the imputed data are generated from the same distribution as the respondents, to other cases of nonignorable missingness. For each

scenario considered by the analyst, we propose calculating the utility measures and the cost for various follow-up sample sizes. As the sample size increases, the utility measures are naturally getting better, but the cost also increases. Thus, the decision rule will depend on the trade-off between these measures. With the values of utility and cost measures for different sample sizes and under different missingness scenarios, the agency can make an informed decision about stopping the data collection.

We describe our approach of adaptive design with imputation methods in Section 2. We describe the methodology used to fit the model to the observed data in Section 2.1, and to generate imputed data sets under MNAR in Section 2.2. We describe the steps to evaluate the adaptive design in Section 2.3. We define the utility and cost measures in Section 2.4 and Section 2.5, respectively. The method is exemplified by an application to the U.S. Census of Manufactures in Section 3. Additional discussion appears in Section 4.

2 Methodology

2.1 Mixture model

Imputation of the missing data is an important part of the adaptive design approach we propose, as we generate completed survey data under different scenarios. These scenarios can include Missing at Random (MAR) data, as well as Missing Not at Random (MNAR). The model has also to be flexible to capture different patterns of the multivariate continuous data, such as multi modality, skewness and correlation. The mixture of multivariate normal distributions is a natural choice, since with a sufficient number of components, it can approximate well any distribution. With a nonparametric prior in a Bayesian framework (Ferguson, 1973, 1983; Escobar and West, 1995; West et al., 1994), we can provide more flexibility and improve the density estimation (Müller and Mitra, 2013).

Let $\boldsymbol{Y} = (\boldsymbol{Y}_1, \ldots, \boldsymbol{Y}_n)'$ denote the p variables of the n respondents. Denote by $z_i \in \{1, \ldots, K\}$ the indicator of which component the i-th observation belongs to with probability $\pi_k = P(z_i = k)$, where $i = 1, \ldots, n$ and $k = 1, \ldots, K < \infty$. Within each component, $\boldsymbol{Y}$ follows a multivariate normal distribution with mean $\boldsymbol{\mu}_k$ and variance Σ_k:

$$\begin{aligned} \boldsymbol{y}_i | z_i, \boldsymbol{\mu}, \boldsymbol{\Sigma} &\sim N(\boldsymbol{\mu}_{z_i}, \Sigma_{z_i}) && (1) \\ z_i | \boldsymbol{\pi} &\sim \text{Multinomial}(\pi_1, \ldots, \pi_K). && (2) \end{aligned}$$

We standardize each dimension of $\boldsymbol{Y}$ to facilitate modeling.There are two alternatives for the prior specification of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$. The first is the conjugate prior, with

$$\begin{aligned} \boldsymbol{\mu}_k | \Sigma_k &\sim N(\boldsymbol{\mu}_0, h^{-1}\Sigma_k) && (3) \\ \Sigma_k &\sim \text{InverseWishart}(f, \Phi), && (4) \end{aligned}$$

where f is the degrees of freedom and $\Phi = diag(\phi_1, \ldots, \phi_p)$ with $\phi_j \sim \text{Gamma}(a_\phi, b_\phi)$ for $j = 1, \ldots, p$. We set $\boldsymbol{\mu}_0 = 0$, since the variables are standardized, $f = p + 1$ to ensure a proper posterior distribution, and $h = 1$ for convenience. The second alternative has the

same prior as (3) for $\boldsymbol{\mu}$, but instead we set the covariance matrices to $\Sigma_k = \sigma I_p$, for all k and for a specified value of $\sigma > 0$. The value of σ controls the tightness of the clusters. In the remaining of this paper, we will use this approach with the fixed covariances, since some preliminary results suggest this approach provides more flexibility for the imputation models.

Following the stick-breaking representation of a truncated Dirichlet process (Sethuraman, 1994; Ishwaran and James, 2001), we define the mixture weights as

$$\begin{aligned} \pi_k &= v_k \prod_{g<k}(1 - v_g) \quad \text{for } k = 1, \ldots, K && (5) \\ v_k &\sim \text{Beta}(1, \alpha) \quad \text{for } k = 1, \ldots, K-1;\ v_K = 1 && (6) \\ \alpha &\sim \text{Gamma}(a_\alpha, b_\alpha). && (7) \end{aligned}$$

The hyperparameters are set following the same specification of Kim et al. (2014), with $a_\alpha = b_\alpha = 0.25$ to allocate the probabilities to the first few components.

2.2 Imputation methods

The model in (1)–(7) can be used to impute values for the missing observations. The imputed values are simulated from the posterior predictive distribution within the Gibbs sampler. If we use the estimated values of $(\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma})$ on the predictive distribution, the imputed values are generated from the same distribution as the observed data when the data are assumed to be MAR. We will use the estimated $(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and specify new mixture probabilities $\boldsymbol{\pi}^*$ to generate imputed values from a distribution different than the observed. This is done when the data are assumed to be MNAR, and $\boldsymbol{\pi}^*$ reflects the assumptions about the missingness pattern.

The model also can be used to fill in the missing data caused by item nonresponse, when some of the variables are observed, and by unit nonresponse, when none of the variables are observed. In both cases, $\boldsymbol{y}$ is sampled from the conditional distribution given the posterior samples of the parameters.

For item nonresponse, we impute the missing variables for each unit from the conditional normal distribution given the observed variables. For this case, we assume that the cause for nonresponse is not related to the missing responses, that is, the item nonresponses are MAR given the observed variables. At each MCMC iteration, after updating $\boldsymbol{\pi}$, $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, we sample the cluster indicators z_i from the multinomial distribution. Then, conditioning on the cluster indicator and on the observed parts, we can impute the missing parts for each incomplete observation from the corresponding conditional normal distribution.

For unit nonresponse, we impute the entire vector of variables for the n_{mis} completely missing respondents from the posterior predictive distribution. In this case, there is no partial information about the location of the responses. Here, we assume that the values are MNAR. We use the samples of $\boldsymbol{\mu}^{(t)}$ and $\boldsymbol{\Sigma}^{(t)}$ from the model fit to the observed data, and specify new mixture probabilities $\boldsymbol{\pi}^*$. For $i = 1, \ldots, n_{\text{mis}}$, we sample

$$\begin{aligned} z_i|\boldsymbol{\pi}^* &\sim \text{Multinomial}(\pi_1^*, \ldots, \pi_K^*) && (8) \\ \tilde{\boldsymbol{y}}_i|z_i, \boldsymbol{\mu}^{(t)}, \boldsymbol{\Sigma}^{(t)} &\sim N(\boldsymbol{\mu}_{z_i}^{(t)}, \Sigma_{z_i}^{(t)}). && (9) \end{aligned}$$

Since we need to identify the components to choose the new probabilities, we propose ranking the clusters post-simulation based on their distance δ to the origin, i.e., $\delta_{\text{orig}} = \boldsymbol{\mu}'\boldsymbol{\mu}$, or to the minimum value with $\delta_{\min} = (\boldsymbol{\mu}-\boldsymbol{y}_{\min})'(\boldsymbol{\mu}-\boldsymbol{y}_{\min})$, where $\boldsymbol{y}_{\min} = (\min_i(\boldsymbol{y}_1), \ldots, \min_i(\boldsymbol{y}_p))$, with $\min_i(\boldsymbol{y}_v) = \min(y_{1v}, \ldots, y_{nv})$ for all variables $v = 1, \ldots, p$. Thus, if we want to impute more data with higher response values, we can inflate the probabilities of the top ranked clusters, and the reverse if we want to generate more data with lower response values. This criterium is defined to facilitate the identification of the clusters, thus it can be defined differently depending on the range of the data.

We also summarize the posterior samples by choosing the MCMC iteration with largest posterior value (Fraley and Raftery, 2007). For each iteration, we evaluate the posterior given the sampled values for the parameters and the sample of the complete observed and imputed $\boldsymbol{Y}$. Then, we select the iteration with maximum posterior value (MAP) to summarize the samples. This is done to simplify the task of setting $\boldsymbol{\pi}^*$ to just one cluster allocation, since this allocation can change from iteration to iteration.

Choosing $\boldsymbol{\pi}^*$ determines which components the nonrespondents are more likely to belong to, and specifies the new pattern for the imputed data. The values in $\boldsymbol{\pi}^*$ are chosen by analyzing the posterior summary of the clusters, and adjusting the values to obtain the desired pattern for the imputed values. For example, analysts can set $\pi_k^* = 0$ for clusters in regions that should not have imputed data according to their beliefs about the the non-ignorable missingness. On the other hand, analysts can increase the probability of clusters where they believe there should be more imputed data than what was observed. The closer the probabilities $\boldsymbol{\pi}^*$ remain to the posterior samples of $\boldsymbol{\pi}$, the closer the data are to MAR.

Specifying the entire vector of probabilities might be complicated in some situations, particularly with large p. To facilitate the process of setting $\boldsymbol{\pi}^*$, we developed the NIMC (Nonignorable missingness Imputation for Multivariate Continuous data) tool. The NIMC is an R (R Core Team, 2013) application developed with the `shiny` package (RStudio and Inc., 2014), where the probabilities can be altered by setting values on sliders for each component. The sliders start with the estimated probabilities of $\boldsymbol{\pi}^{(t)}$ for a given iteration t as default values, such as the MAP iteration, and the user selects the factors to be applied to each probability. The slider values are renormed to sum to one so the resulting mixture model in (**??**) is a proper density. Using the renormed probabilities, the application automatically generates synthetic data following the model described in (8)–(9), so that the analyst can change $\boldsymbol{\pi}^*$ to get the desired distributions.

The application presents some tabs with the results of the imputation. The main tab includes pairwise scatterplots of the imputed and observed data, and the 95% quantile ellipses of the fitted clusters. We include the plot with the log transformed and standardized data, on the scale in which the model was fit, and the plot with the data on its original scale. For the latter, the user has an option to choose between the raw data and the data with the log transformation to help visualize skewed distributions. To help with the visual aspect of the scenario assessment, a second tab includes summary statistics for the observed and imputed data separately, and the merged completed data set. The summary statistics are also calculated for the standardized and original data. The user can select the summary

statistics, depending on the variables being analyzed. These measures, in combination with the scatterplots, provide different perspectives to help the analyst make decisions about the sensitivity analysis. As a baseline, we include correlations between the response variables and descriptive statistics for the marginal variables, with quantiles, means, minimum values and maximum values. Finally, when the imputation results are satisfactory, the completed data set on the original scale can be visualized and downloaded in the third tab.

2.3 Adaptive Design

Let us consider an ongoing survey which is going to be evaluated at a certain point in time. The goal is to decide if it is worth spending any more resources collecting more data, or if what has been collected so far is adequate to enable inferences for population quantities. This decision is based on the trade-off between the improvement in the data representativeness and the increase in cost of collecting more data.

We denote by N the intended size of the entire sample. After the first wave of data collection, this sample is partitioned into the respondents, who have at least one of the variables recorded, and the nonrespondents, who did not provide any response to the survey yet. To deal with the item nonresponse among the respondents, we follow the first approach described in Section 2.2 and impute the missing values assuming MAR conditioned on the observed values.

We consider two options to obtain a completed data set of size N. The first option is to stop collecting data and impute the missing data based on the information available from the observed data. We fit the multivariate mixture model described in Section 2.1 to the n_R respondents denoted by D_R. The remaining n_{NR} nonrespondents in set D_{NR} are then imputed based on this model, as seen in Figure 1. The imputation at this stage can be done under scenarios of either MAR or MNAR, depending on the purpose of the completed data. When generating a completed data set to be released for general analysis, the missing data are probably going to be imputed assuming MAR. When doing sensitivity analysis to evaluate the impact of missingness patterns, we consider different possible scenarios, including MNAR. We show some different scenarios in the results in Section 3.

The second option is to collect n_F more observations to form a follow-up sample, D_F. The set of n_{NF} cases that are still missing after the follow-up sample are denoted by D_{NF}. The mixture model is fit again, either to all the observations in $(D_R + D_F)$ or just D_F, as seen in Figure 2(a) and 2(b) respectively. These alternatives correspond to the belief that the nonrespondents in D_{NF} are more similar to the entire set of observations or just to the latest wave. It may be beneficial to use $(D_R + D_F)$ when n_F is too small to support reliable modeling.

The choice in Figure 2(a) is appropriate if D_R and D_{NR} come from the same distribution and the latter is MAR. Here, D_F is a random sample of the nonrespondents, and D_{NF} should be imputed from the same distribution as all the data that was observed so far. It also makes sense to follow 2(a) if D_R and D_{NR} have different distributions, but D_F comes from the same distribution of D_R, and the differences in the distributions are reflected on a MNAR imputation model. The choice in Figure 2(b) makes more sense if it is believed that D_R and

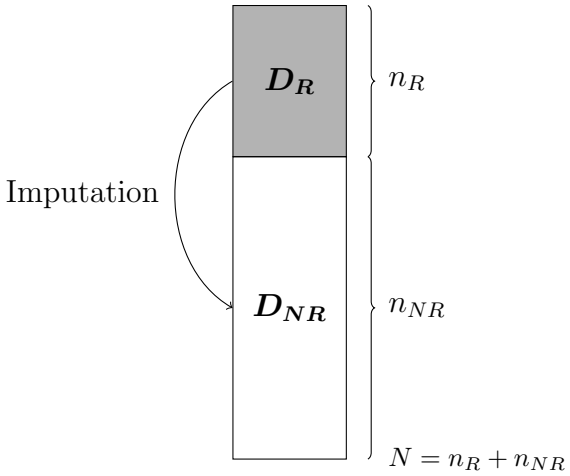


Figure 1: Imputation diagram if decide to stop data collection

D_{NR} come from different distributions, D_F is a random sample of the nonrespondents, and D_{NF} should be imputed based only on the distribution of the follow-up sample.

The decision between the two options represented in Figures 1 and 2 depends on the trade-off between utility and cost of collecting more data. In order to evaluate the utility of what has been collected so far, we propose comparing some utility measures for different follow-up sample sizes and under different missingness scenarios via sensitivity analysis.

With the data collected on the first survey wave, we fit the multivariate mixture model from Section 2.1 to the observed values. After the convergence of the MCMC and the selection of the iteration with maximum posterior value, we summarize the model estimate with the selected MAP posterior samples of the mixture parameters. The imputed values due to item nonresponse are also selected from the MAP iteration. Combined with the actual recorded values, they form the set D_R that is going to be used in all the imputation scenarios.

With the fitted cluster allocation, the next step is to consider some plausible missingness scenarios for sensitivity analysis. This is done by specifying new probabilities $\boldsymbol{\pi}^*$ to reflect different distributional patterns for the nonrespondents. To facilitate this step, we developed the NIMC (Nonignorable missingness Imputation for Multivariate Continuous data) application. The NIMC application provides an interface where the user can set the values of $\boldsymbol{\pi}^*$ with sliders for each component, and immediately see pairwise scatterplots and summary statistics of the observed and imputed data.

The scenarios can include the MAR case as a baseline, as well as other MNAR cases that are reasonable for the survey context. An example of a MNAR case is when the probabilities in $\boldsymbol{\pi}^*$ are increased for the top ranked clusters to reflect a larger proportion of missing data with larger responses.

These scenarios are used to compare the impact on inferences of different sizes of follow-up samples, starting from what has been observed so far, and progressing as the follow-up

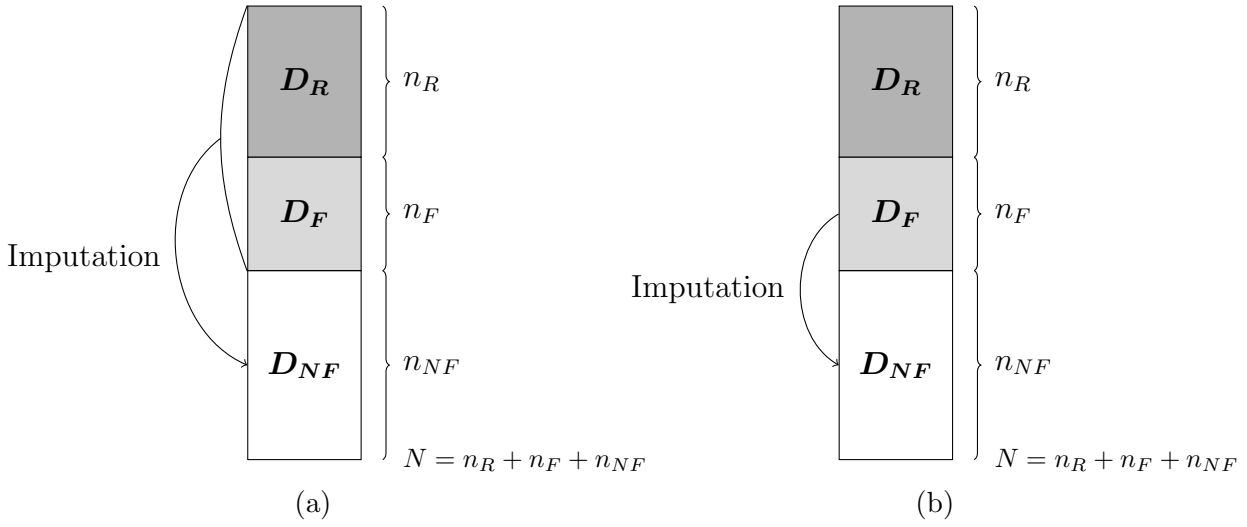


Figure 2: Imputation diagrams if decide to collect follow-up sample

sample size increases. As n_F increases, inferences on the MAR case are not expected to change as much as on the MNAR case. If the impact on the inferences is significant, the agency might decide that it is worth collecting more data. If even in an extreme MNAR scenario the impact is not significant, then the agency might decide to stop data collection.

We describe now the data sets that will be used to evaluate these impacts. Define n_{MAX} as the maximum sample size that can be collected given a total budget or a time restriction. We consider sampling different proportions of the maximum sample size, such that $n_F = \delta \times n_{MAX}$ where n_F is the follow-up sample size and $\delta \in [0, 1]$. Define m_P as the number of multiple imputations we generate when imputing to create the "population" according to the missingness scenario, and m_F as the number of multiple imputations we generate when imputing the remaining data after the follow-up sample.

We denote by $\Omega = (\boldsymbol{\pi}^{*(1)}, \ldots, \boldsymbol{\pi}^{*(S)})$ the set of probabilities of each scenario to be considered. For each value of $\boldsymbol{\pi}^{*(s)} \in \Omega$, with $s = 1, \ldots, S$, we do the following procedure:

1. Generate m_P completed hypothetical populations $(P^{(s,1)}, \ldots, P^{(s,m_P)})$ by multiply imputing all the non-respondents with the fitted model and the probabilities $\boldsymbol{\pi}^{*(s)}$. The observed values D_R are common to all s. The imputed values for each hypothetical population are denoted by $\tilde{D}_{NR}^{(s,j)}$, such that $P^{(s,j)} = D_R \cup \tilde{D}_{NR}^{(s,j)}$, for $j = 1, \ldots, m_P$.

2. For each value of δ under consideration, and for each $j = 1, \ldots, m_P$:

   (a) Obtain a random sample of size n_F from each $\tilde{D}_{NR}^{(s,j)}$. Denote this follow-up sample set by $D_{F,\delta}^{(s,j)}$.

   (b) Fit new mixture models to one or both:

      i. $D_R \cup D_{F,\delta}^{(s,j)}$

ii. $D_{F,\delta}^{(s,j)}$

depending on the option selected from Figure 2(a) and Figure 2(b).

(c) Based on the posterior samples of the new mixture models and assuming MAR, generate m_F multiply imputed values for the n_{NF} observations that are still missing. The imputed cases are denoted by $\tilde{D}_{NF,\delta}^{(s,j,l)}$, with $l = 1, \ldots, m_F$. The completed data sets are denoted by $\tilde{D}_{\delta}^{(s,j,l)} = D_R \cup D_{F,\delta}^{(s,j)} \cup \tilde{D}_{NF,\delta}^{(s,j,l)}$.

(d) Compare the sets $P^{(s,j)}$ and $\left(\tilde{D}_{\delta}^{(s,j,1)}, \ldots, \tilde{D}_{\delta}^{(s,j,m_F)}\right)$ with the utility measures described next.

We generate these multiple draws to account for the uncertainty due to the missing data on each stage of the process. The uncertainty from selecting the follow-up sample could also be considered with repeated sampling of $D_{F,\delta}^{(s,j)}$. However, this source of uncertainty is already accounted for with a reasonable number m_P of multiple imputations.

2.4 Utility measures

The decision of stopping the data collection depends critically on the amount of information that has been collected so far. This information can be measured by the difference in inferences based on what was observed and inferences with a larger sample, under different missingness hypothesis. If the observed sample size is small and the non-respondents have an extremely different distribution than what was recorded, then the impact of a follow-up sample on the estimates is going to be bigger than if the initial sample was larger or the non-respondents were closer to be missing at random.

Consider the two complete data sets generated in Section 2.3: $P^{(s,j)}$, the hypothetical population for scenario s and imputation j; and each $\tilde{D}_{\delta}^{(s,j,l)}$, the completed data set considering the follow-up sample for imputation l. We consider the following measures to compare the similarities between $P^{(s,j)}$ and $\tilde{D}_{\delta}^{(s,j,l)}$.

2.4.1 Measure τ:

Let $\bar{Y}_v^{P(s,j)} = \sum_{i=1}^{N} Y_{i,v}^{P(s,j)}/N$ denote the marginal mean of each variable $v = 1, \ldots, p$ computed from the data set $P^{(s,j)}$, and let $\bar{Y}_v^{D,\delta(s,j,l)} = \sum_{i=1}^{N} Y_{i,v}^{D,\delta(s,j,l)}/N$ denote the marginal mean of each variable $v = 1, \ldots, p$ computed from the data set $\tilde{D}_{\delta}^{(s,j,l)}$. Similarly, let $\hat{\sigma}_v^{P(s,j)}$ and $\hat{\sigma}_v^{D,\delta(s,j,l)}$ denote the standard deviations for each variable v from the data sets $P^{(s,j)}$ and $\tilde{D}_{\delta}^{(s,j,l)}$, respectively. For each variable $v = 1, \ldots, p$, compute

$$t_v^{\delta(s,j,l)} = \frac{\left(\bar{Y}_v^{P(s,j)} - \bar{Y}_v^{D,\delta(s,j,l)}\right)}{\sqrt{\frac{\left[\left(\hat{\sigma}_v^{P(s,j)}\right)^2 + \left(\hat{\sigma}_v^{D,\delta(s,j,l)}\right)^2\right]/2}{N}}}. \tag{10}$$

Let the summary measure be

$$\tau^{\delta(s,j,l)} = \frac{1}{p}\sum_{v=1}^{p}\left|t_v^{\delta(s,j,l)}\right|, \tag{11}$$

for each value of δ, scenario s, population j, and imputation l.

2.4.2 Measure θ:

We also define a measure based on the Mean Absolute Percentage Error, a measure used to measure the percentage error between forecasts and observed values. Here, we consider the average percentage difference between $\bar{Y}_v^{P(s,j)}$ and $\bar{Y}_v^{D,\delta(s,j,l)}$, with respect to $\bar{Y}_v^{P(s,j)}$. Thus, the summary measure is

$$\theta^{\delta(s,j,l)} = \frac{1}{p}\sum_{v=1}^{p}\left|\frac{\bar{Y}_v^{P(s,j)} - \bar{Y}_v^{D,\delta(s,j,l)}}{\bar{Y}_v^{P(s,j)}}\right|, \tag{12}$$

for each value of δ, scenario s, population j, and imputation l.

2.4.3 Measure ρ:

Since these measures are focused on the marginal differences, characteristics of the multivariate data may not be captured. Woo et al. (2009) propose global measures of data utility to compare multivariate distributions of two data sets, in an attempt to quantify the dissimilarity between a masked data set and an original confidential data set that cannot be released. They found that a measure based on propensity scores is the most promising to reflect characteristics of the entire distribution for different types of data, while being computationally feasible to implement.

Propensity scores are used in observational studies for matching covariate characteristics and reduce the impact of confounding factors between groups when inferring treatment effects. The propensity score is defined as $e(\boldsymbol{x}) = P(T = 1|\boldsymbol{x})$, the probability of being assigned to be on the treatment group T given the variables $\boldsymbol{x}$. Woo et al. (2009) propose calculating the propensity score on the merged data set consisting of the original and the masked data. The treatment response T is the indicator for the synthetic data, and the propensity score is estimated for each unit in the merged data set. The similarity between the two groups is assessed by the distribution of the estimated propensity scores and how close they are to 0.5, the reference for indistinguishable groups.

The model specified to estimate the propensity scores has a great impact on this measure. The common approach is to use simple logistic regression of the treatment indicator on the covariates, which assumes a linear relationship between the link function and the covariates. Woo et al. (2008) propose a more flexible approach by using generalized additive models (GAM) instead of the standard logistic regression. The linear component of the regression is replaced by a flexible additive function:

$$\text{logit}\Big(e(\boldsymbol{x})\Big) = \log\left(\frac{e(\boldsymbol{x})}{1-e(\boldsymbol{x})}\right) = \alpha + f_1(x_1) + \cdots + f_p(x_p), \tag{13}$$

where each $f_j(x_j)$ is a smooth function of x_j, for example regression splines. This model outperforms the logistic regression and facilitates the modeling of non-liner relationships. However, as in the logistic regression, the results are dependent on the variables and interactions that are included in the model. The recommendation remains to include the covariates believed to be related to the "treatment" indicator.

We define the measure ρ based on the propensity score (Woo et al., 2009) estimated with GAM (Woo et al., 2008), with the covariates as all the response variables $\boldsymbol{y} = (y_1, \ldots, y_p)$ included as main effects. The two data sets to be compared, $P^{(s,j)}$ and $\tilde{D}_{\delta}^{(s,j,l)}$, are merged together. The set $P^{(s,j)}$ receives an indicator variable $T = 1$, while $\tilde{D}_{\delta}^{(s,j,l)}$ has $T = 0$. The logistic model is fit on the response variable T, for the merged data set of size $2N$. The predicted values of each observation $\hat{e}_i$ should be around 0.5 if the two data sets are comparable. The overall measure is given by

$$\rho^{\delta(s,j,l)} = \frac{\sum_{i=1}^{2N}(\hat{e}_i - 0.5)^2}{2N}, \tag{14}$$

for each value of δ, scenario s, population j, and imputation l. If the distributions are equal, ρ is close to zero. On the other extreme, if the two distributions are very distinct, then $\rho \sim 1/4$.

As n_F increases, the two data sets intersect more and, obviously, all the measures will decrease. Therefore, the interest is on their relative values compared between the different follow-up sample sizes and for the different scenarios. Together with the cost measures, the agency conducting the survey can make a decision about stopping data collection.

2.5 Cost measure and decision rule

Together with the utility measures described above, with a measure cost, one can make the decision about the follow-up sample and its size. We propose obtaining the results with a set of different values of n_F, calculating the utility measures (τ, θ, ρ) and estimating the cost of collecting the additional sample. With this information, the agency can decide how much it is willing to spend for the utility improvements. The values can be explored with a table or plot, as we show in Section 3.

Denote by C_F the total cost of the follow-up sample, C_0 the fixed cost regardless of the sample size, and c the cost of selecting, measuring and processing each of the n_F follow-up sample units. For our purposes of creating a decision rule for adaptive design, we consider sufficient to estimate the cost with a linear function of the sample size. More complex cost functions are discussed in Groves (2004). The total cost is estimated by

$$C_F = C_0 + (c \times n_F). \tag{15}$$

We note that the maximum sample size n_{MAX}, defined and used in Section 2.3, can be obtained by solving for n_F in (15) given the total budget available.

3 Illustration with Census of Manufactures Data

We demonstrate now the proposed approach using data from the 2007 U.S. Census of Manufactures (CMF). The original data consist of responses collected from forms sent to companies representing all U.S. locations and manufacturing industries. The form queries include information about sales, employees and payroll of the businesses. As in many other large scale surveys, the CMF faces the problem of missing data, from both item and unit nonresponse. Because of this, the U.S. Census Bureau spends resources trying to reduce the nonresponse rates during data collection, for example, by resending the forms, or allocating Census consultants to work specifically with some of the largest companies. Thus, the CMF could potentially benefit from an adaptive survey design to stop data collection earlier. This would not only reduce the cost of the survey, but it can also be beneficial for users with data released sooner.

The CMF was the main motivating case for the proposed adaptive design methodology with nonignorable missingness. This is because the data are collected and processed in waves during the year, depending on when each firm sends its form. With this temporal pattern, it is natural to think about evaluating the survey design at certain time points, e.g., monthly or quarterly. This pattern is also a mechanism that can lead to missing not at random data. It can happen, for example, if smaller companies tend to not send their forms on time. If that is the case, with our imputation model, we can inflate the probabilities of bottom-ranked clusters to impute more data with lower response values. This and some other scenarios are reasonable assumptions to consider when making sensitivity analysis, as we demonstrate here.

In order to investigate the missingness patterns and apply our imputation method, we focus on some variables of the CMF. The variables are total value of shipments (TVS), total employment (TE), and salary/wages (SW). The selected industry is ready-mix concrete manufacturing, which is a homogeneous industry since most of their business is based on a single product.

For illustration, we consider as observed data all the units that had at least one reported value among the three variables of interest and that had a valid response date. Some observations met the first criterion, but there was no response date available on file for miscellaneous reasons, which could include the case of late respondents that sent their information after some processing deadline. We consider these cases as missing data, and impute their entire vector of responses after fitting the mixture model in (1)–(7) to the observed data.

We selected three scenarios to consider for the sensitivity analysis. The first is the MAR case, with the estimated probabilities $\boldsymbol{\pi}$, as a baseline. The second is the MNAR case with higher probabilities for bottom ranked clusters, to generate small response values. The third scenario is the opposite MNAR case, with higher probabilities for top ranked clusters. This is included in the sensitivity analysis as the worst case scenario, since larger companies will have more impact in some summary statistics. With the NIMC application, we specified the values of the probabilities and obtained $\Omega = (\boldsymbol{\pi}^{*(\text{MAR})}, \boldsymbol{\pi}^{*(\text{bottom})}, \boldsymbol{\pi}^{*(\text{top})})$ corresponding to three scenarios described above. The values of $\boldsymbol{\pi}^*$ for each scenario can be seen in the legends of Figure 3, Figure 4 and Figure 5.

We considered $\delta = (0, 0.25, 0.5, 0.75, 1)$, that is, collecting follow-up samples with 0%, 25%, 50%, 75% and 100% of n_{MAX}. For this example, we set $n_{MAX} = n_{NR}$, assuming the budget and time available are enough to attempt to collect all nonrespondents. For each scenario, first we created $m_P = 10$ complete hypothetical populations. Then, for each population $P^{(s,j)}$ and for each value of δ, we obtained the follow-up sample $D_{F,\delta}^{(s,j)}$. We fit two new mixture models to: (i) the observed data set and the follow-up sample $\left(D_R \cup D_{F,\delta}^{(s,j)}\right)$; and to (ii) the follow-up sample only $\left(D_{F,\delta}^{(s,j)}\right)$. With the posterior samples from the new mixture models, we create $m_F = 5$ complete data sets through multiple imputation of the still remaining nonrespondents. For all scenarios, we calculated the utility measures ρ, τ and θ as described in Section 2.4. Each measure is summarized by the mean and standard deviation of the individual values from m_P population repetitions and the m_F multiple imputations.

We do not include values for the cost measure due to disclosure limitations of the CMF data, per request of the Census Bureau. The cost measure in (15) is a linear function of n_F, and thus, a linear function of δ. Thus, we believe it is sufficient to present the results based on δ for the purpose of sensitivity analysis and comparison of the different scenarios.

In Figure 3, we see the pairwise scatterplots of the results of the MAR scenario for the concrete industry. The axes of all the scatterplots were removed to prevent disclosure of information about the magnitude of the data. The mixture model resulted in 17 nonempty components plotted by the colored circles in the upper diagonal plots. From the estimated probabilities in the legend, we can see that most of the weight is concentrated in less than half of the clusters. The black points in the lower diagonal plots are imputed assuming MAR. In Figure 4, we can see the imputation results of the second scenario, with higher probabilities for bottom ranked clusters. Because of this pattern on the probabilities, there are more imputed points on the lower tails of the data. In Figure 5, we can see the imputation results of the opposite scenario, with higher probabilities for the top ranked clusters.

In Table 1, we can see the utility measures of the MAR scenario for the concrete industry, with the two approaches for the new mixture model. In Table 2, we have the results for the MNAR scenario with higher probabilities for bottom ranked clusters, and in Table 3, we have the results for the MNAR scenario with higher probabilities for top ranked clusters. As expected, in all cases the measures decrease as δ increases. When $\delta = 1$, we sample all the units and there is no need to fit the model again and impute more data. Thus, the two populations being compared, $P^{(s,j)}$ and $\tilde{D}_{\delta}^{(s,j,l)}$, are the same and the utility measures are equal to zero. When $\delta = 0$, we do not collect any more data, so there is no follow-up sample to fit the model on the second approach.

The MAR scenario is the baseline and the measures are relatively small, since the two data sets being compared should come from very similar distributions. In the MNAR scenarios, the values from the model fit with just the follow-up sample in Table 2(b) and Table 3(b) are smaller than the values from the model fit with the observed data. This is expected, since the distribution of the nonrespondents of the first wave, plotted as the black points in Figure 4 and Figure 5, is very different than the distribution of the respondents, plotted as the gray points. The second approach, therefore, creates data sets $\tilde{D}_{\delta}^{(s,j,l)}$ that are more

similar to the populations $P^{(s,j)}$.

The measures are also compared with the plots in Figure 6. In these plots, we can compare the decrease of the measures between the different scenarios. The model fit with the observed data and the follow-up sample, plotted on the left column, resulted in very distinct measures for the scenarios. On the right column, the model fit with just the follow-up sample resulted in overlapping measures that are much smaller than with the first option. This confirms that the two options are generating different completed data sets, and that using the model with just the follow-up sample is better when the data are MNAR. The values of θ, the measure based on the mean absolute percentage error, with the model fit with just the follow-up sample are larger for the MAR scenario than for the other scenarios. This could be caused by the instability of the percentage error when the values are close to zero, one of the drawbacks of the mean absolute percentage error. If the data are MAR, there is not much gain with a follow-up sample. If the data are MNAR, there is a significant utility improvement in collecting 25-50% more data. Beyond that, the utility increase might be not worth the increase in the cost.

With the utility measures and the cost of the follow-up samples, the agency should be able to make a decision about the survey. However, we can also make a decision based on a more formal method. As an example, let us consider the sample size that minimizes the sum of variance plus cost, as suggested by Groves (2004). In our case, we replace the variance by one of the utility measures, so that we minimize the measure of how far the imputed data are from the population. We choose the measure $\bar{\rho}$ to capture this overall distance between the data sets better, since the other measures are based on the distance between the marginal means. For the cost, we will use δ.

To make the variance and cost be on the same scale, Groves (2004) transforms the cost by multiplying it by a constant λ. Since in our illustrative example the proxy for cost, δ, is already on a fixed scale from 0 to 1, we will transform the values of $\bar{\rho}$ to be on the same scale by subtracting the minimum and dividing by the maximum at each scenario. Then, we select the follow-up sample size that minimizes the sum of this transformed $\bar{\rho}$ and δ. These cases are highlighted in bold in Table 1, Table 2 and Table 3. We note that under this transformation, the extreme cases of $\delta = 0$ and $\delta = 1$ are equivalent. The agency can consider other decision rules to guide the next stages of the survey.

Table 1: Summary of utility measures with the results for the Concrete industry from the 2007 CMF, for the MAR imputation scenario. The results highlighted in bold correspond to the follow-up sample size that minimized the sum of cost and the transformed measure $\bar{\rho}$. *The values of $\bar{\rho}$ (sd) should be multiplied by 10^{-5}.

(a) New mixture model fit to observed data and follow-up sample $\left(D_R \cup D_{F,\delta}^{(\text{MAR})}\right)$

δ	$\bar{\rho}$ $(sd)^*$	$\bar{\tau}$ (sd)	$\bar{\theta}$ (sd)
0	3.465 (1.823)	0.372 (0.242)	0.255 (0.229)
0.25	**2.514 (1.258)**	**0.290 (0.139)**	**0.185 (0.110)**
0.5	1.970 (1.045)	0.235 (0.119)	0.139 (0.082)
0.75	0.886 (0.348)	0.144 (0.091)	0.094 (0.075)
1	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)

(b) New mixture model fit to follow-up sample only $\left(D_{F,\delta}^{(\text{MAR})}\right)$

δ	$\bar{\rho}$ $(sd)^*$	$\bar{\tau}$ (sd)	$\bar{\theta}$ (sd)
0	–	–	–
0.25	6.858 (3.299)	0.424 (0.273)	0.279 (0.183)
0.5	3.662 (1.547)	0.313 (0.144)	0.224 (0.123)
0.75	**1.251 (0.573)**	**0.188 (0.092)**	**0.130 (0.074)**
1	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)

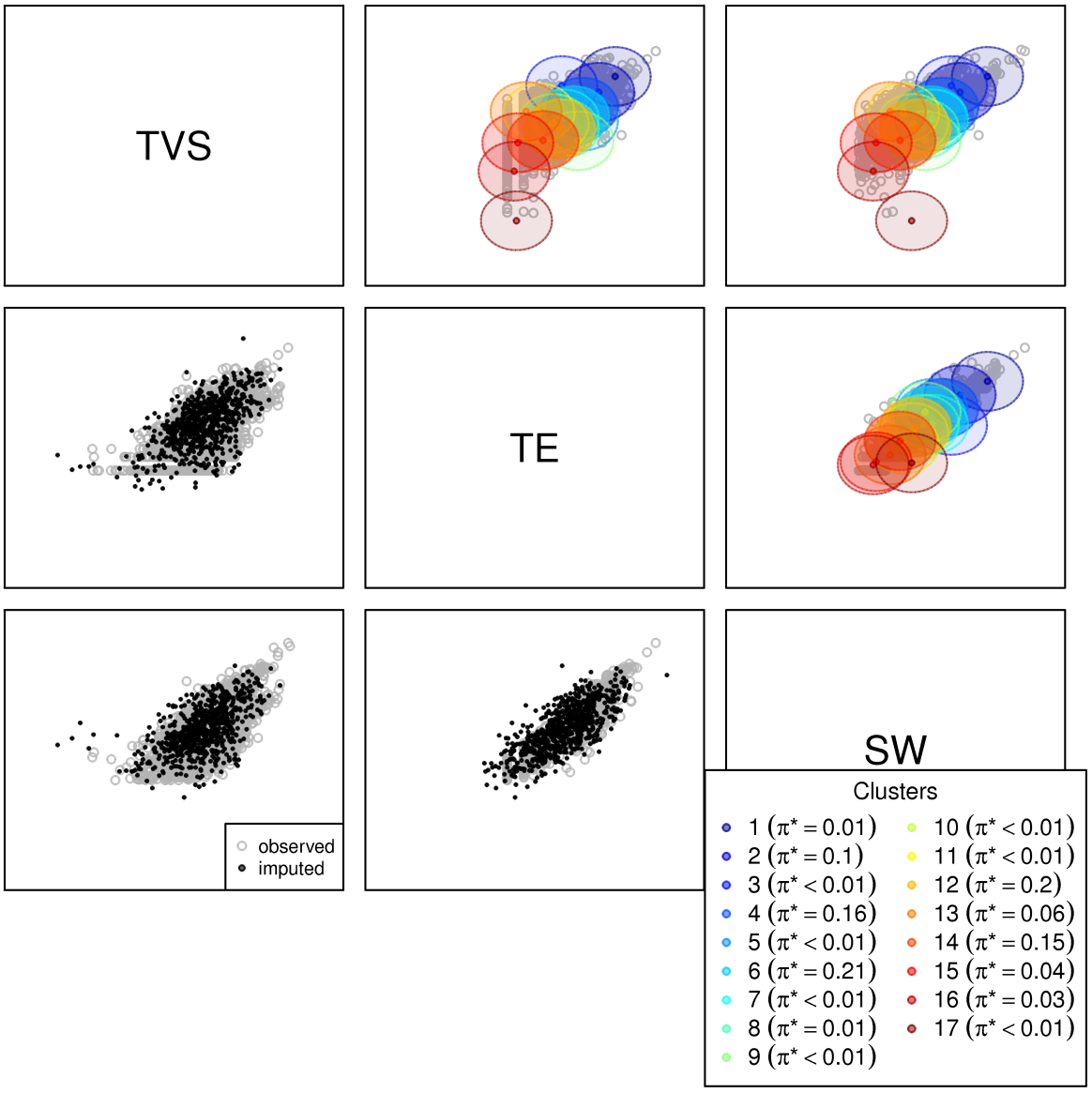


Figure 3: Pairwise scatterplots with the results for the Concrete industry from the 2007 CMF, for the MAR imputation scenario. Observed points are plotted as gray hollow circles. The black filled circles on the lower diagonal are the imputed points. The colored circles on the upper diagonal are the 95% quantile ellipses of the fitted clusters, with color intensity proportional to the mixture probabilities.

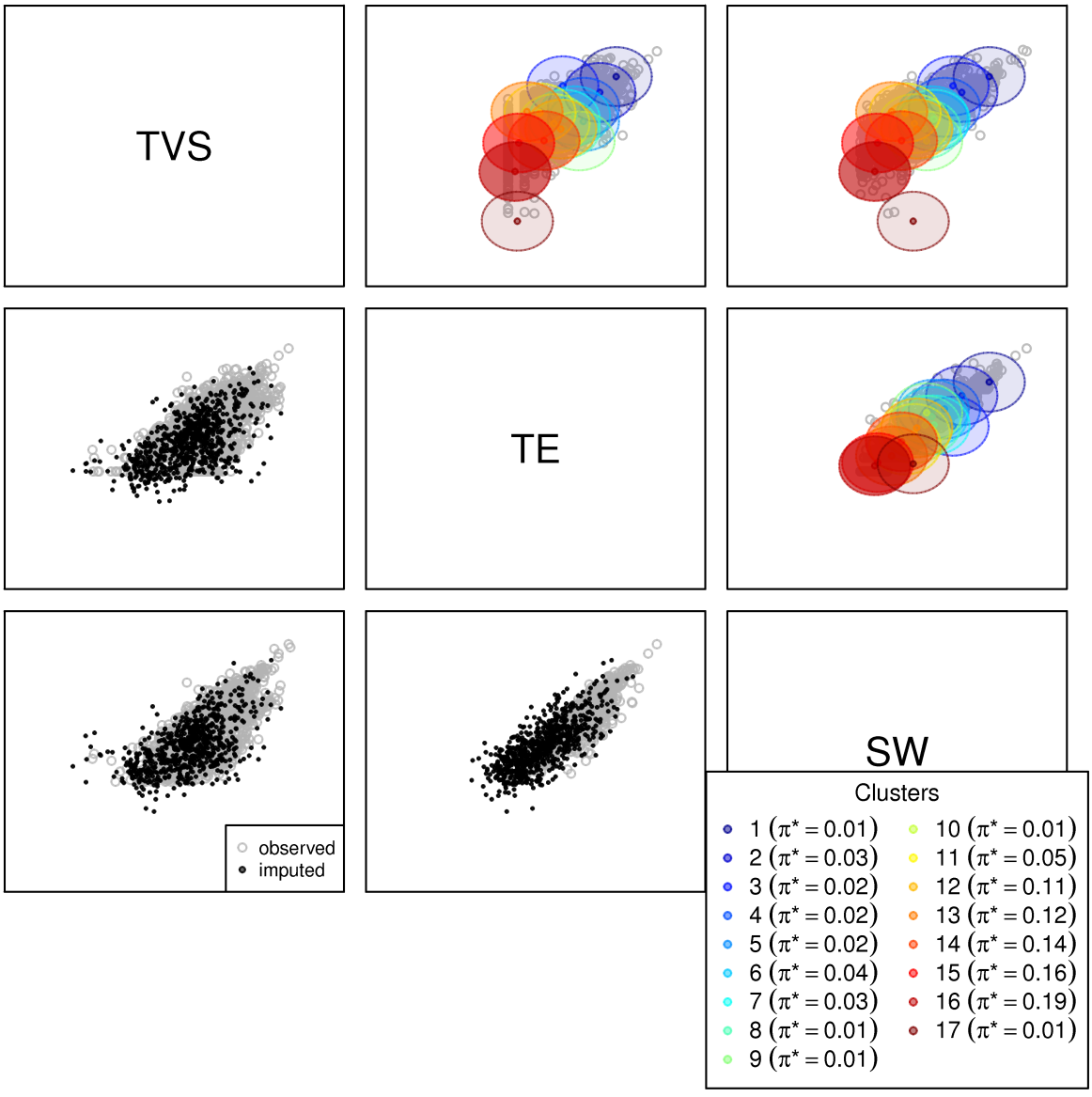


Figure 4: Pairwise scatterplots with the results for the Concrete industry from the 2007 CMF, for the MNAR imputation scenario with higher probabilities for bottom ranked clusters. Observed points are plotted as gray hollow circles. The black filled circles on the lower diagonal are the imputed points. The colored circles on the upper diagonal are the 95% quantile ellipses of the fitted clusters, with color intensity proportional to the mixture probabilities.

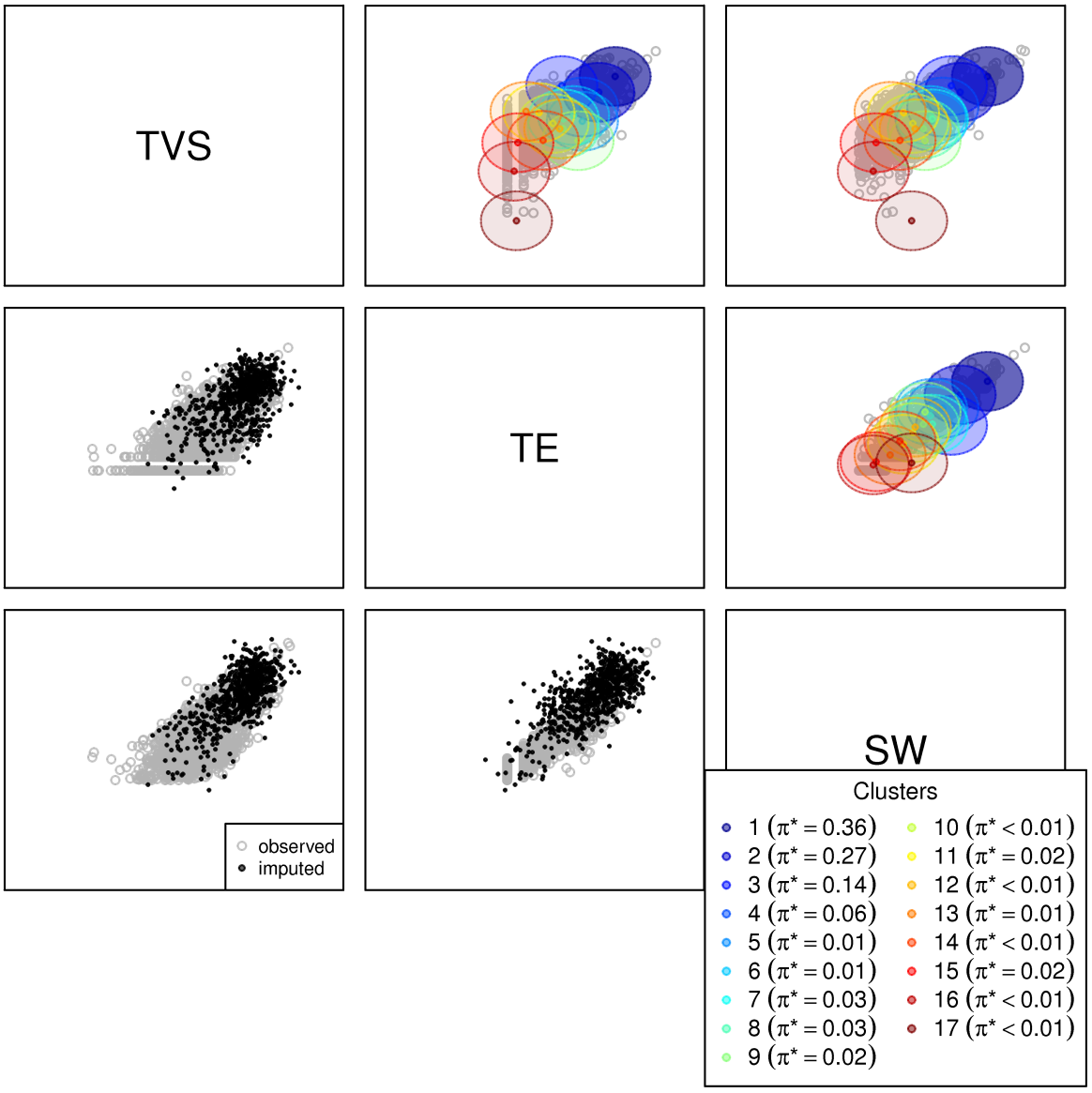


Figure 5: Pairwise scatterplots with the results for the Concrete industry from the 2007 CMF, for the MNAR imputation scenario with higher probabilities for top ranked clusters. Observed points are plotted as gray hollow circles. The black filled circles on the lower diagonal are the imputed points. The colored circles on the upper diagonal are the 95% quantile ellipses of the fitted clusters, with color intensity proportional to the mixture probabilities.

Table 2: Summary of utility measures with the results for the Concrete industry from the 2007 CMF, for the MNAR imputation scenario with higher probabilities for bottom ranked clusters. The results highlighted in bold correspond to the follow-up sample size that minimized the sum of cost and the transformed measure $\bar{\rho}$. *The values of $\bar{\rho}$ (sd) should be multiplied by 10^{-5}.

(a) New mixture model fit to observed data and follow-up sample $\left(D_R \cup D_{F,\delta}^{(\text{bottom})}\right)$

δ	$\bar{\rho}$ $(sd)^*$	$\bar{\tau}$ (sd)	$\bar{\theta}$ (sd)
0	117.986 (12.423)	5.395 (0.396)	0.958 (0.046)
0.25	60.020 (6.602)	3.866 (0.273)	0.691 (0.038)
0.5	**24.628 (4.492)**	**2.401 (0.275)**	**0.422 (0.039)**
0.75	6.260 (1.712)	1.231 (0.184)	0.220 (0.031)
1	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)

(b) New mixture model fit to follow-up sample only $\left(D_{F,\delta}^{(\text{bottom})}\right)$

δ	$\bar{\rho}$ $(sd)^*$	$\bar{\tau}$ (sd)	$\bar{\theta}$ (sd)
0	–	–	–
0.25	12.833 (5.391)	0.642 (0.284)	0.122 (0.063)
0.5	**3.828 (1.896)**	**0.312 (0.136)**	**0.059 (0.026)**
0.75	1.460 (0.661)	0.201 (0.111)	0.037 (0.023)
1	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)

Table 3: Summary of utility measures with the results for the Concrete industry from the 2007 CMF, for the MNAR imputation scenario with higher probabilities for top ranked clusters. The results highlighted in bold correspond to the follow-up sample size that minimized the sum of cost and the transformed measure $\bar{\rho}$. *The values of $\bar{\rho}$ (sd) should be multiplied by 10^{-5}.

(a) New mixture model fit to observed data and follow-up sample $\left(D_R \cup D_{F,\delta}^{(\text{top})}\right)$

δ	$\bar{\rho}$ $(sd)^*$		$\bar{\tau}$ (sd)		$\bar{\theta}$ (sd)	
0	411.057	(19.222)	8.732	(0.285)	1.415	(0.049)
0.25	197.266	(12.833)	6.278	(0.269)	1.026	(0.037)
0.5	**80.489**	**(12.273)**	**4.057**	**(0.282)**	**0.668**	**(0.046)**
0.75	16.758	(2.476)	1.910	(0.155)	0.315	(0.026)
1	0.000	(0.000)	0.000	(0.000)	0.000	(0.000)

(b) New mixture model fit to follow-up sample only $\left(D_{F,\delta}^{(\text{top})}\right)$

δ	$\bar{\rho}$ $(sd)^*$		$\bar{\tau}$ (sd)		$\bar{\theta}$ (sd)	
0	–		–		–	
0.25	12.567	(7.297)	0.541	(0.271)	0.088	(0.049)
0.5	4.731	(2.617)	0.311	(0.177)	0.054	(0.030)
0.75	**1.451**	**(0.820)**	**0.171**	**(0.092)**	**0.029**	**(0.017)**
1	0.000	(0.000)	0.000	(0.000)	0.000	(0.000)

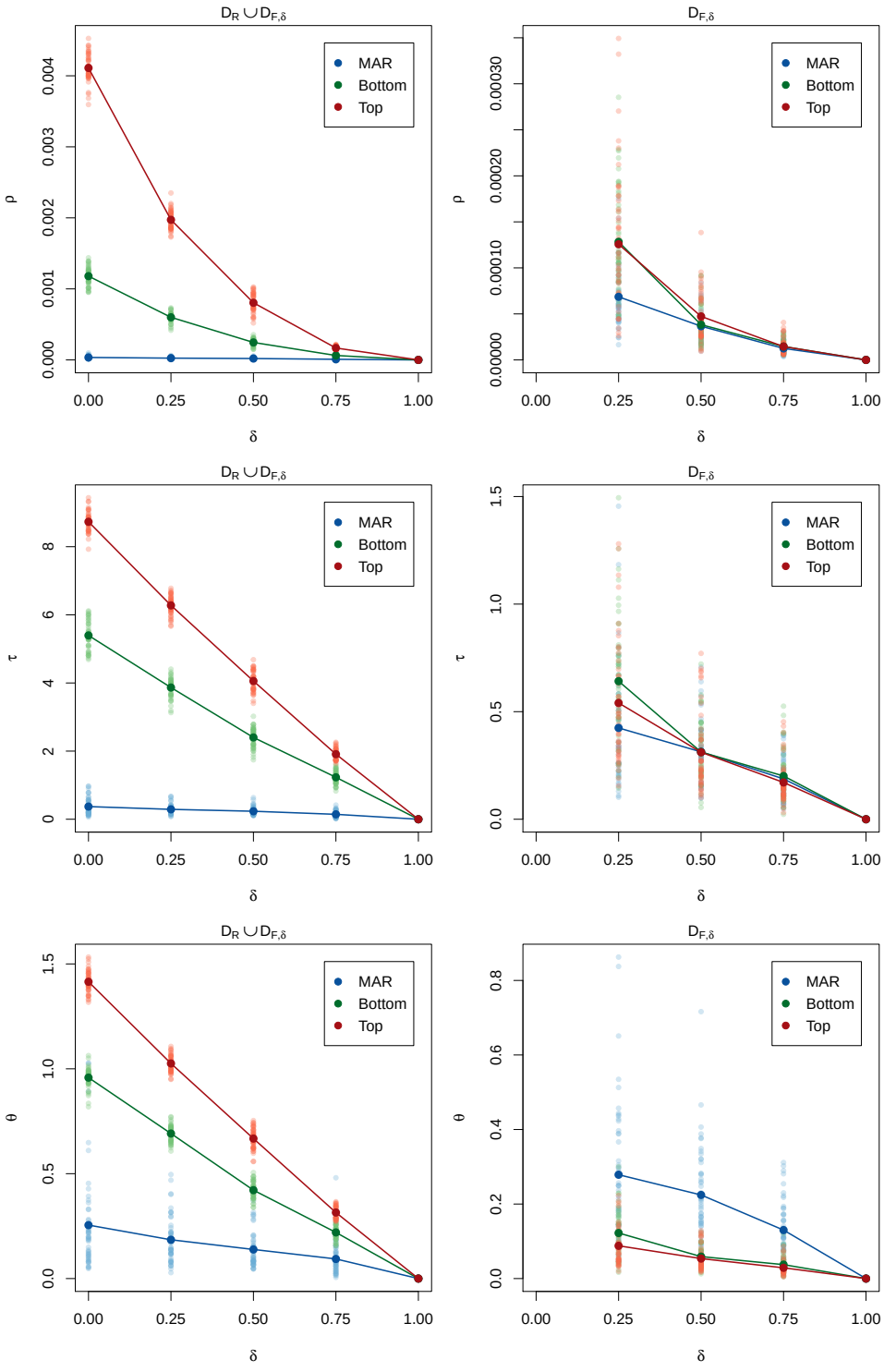


Figure 6: Summary of utility measures for the three scenarios considered for the Concrete industry from the 2007 CMF. The results for ρ, τ and θ are on the first, second and last row, respectively. The plots on the right column contain the results with the model fit to observed data and follow-up sample, while the plots on the right contain the results with the model fit to follow-up sample only. The faded points are the individual values for each multiple imputation.

The results confirm that when the data are missing at random, the best results are obtained with the observed data set and follow-up sample. If the nonrespondents are missing not at random, it is better to use the second proposed approach: if collecting a follow-up sample, use this to fit a new mixture model and impute the remaining data. In the MNAR cases, we can see from the tables and plots for both industries that collecting a follow-up sample of 25%-50% of the nonrespondents, can result in utility measures similar to the MAR case. Comparing these values and considering the linear cost increase alongside with the increase in the follow-up sample size, the agency can make their decision about how to proceed with data collection.

4 Conclusions

We present an approach for adaptive survey designs using methods for imputation of multivariate continuous data with nonignorable missingness. During an ongoing survey, the agency can decide to stop data collection or obtain follow-up samples based on the information that has been collected so far. This decision is based on the representativeness of the respondents, and how the nonrespondents can affect the inference results. In the case of interrupting data collection and imputing the missing data, we can reduce costs and allocate resources more efficiently.

We also propose an imputation method, which estimates the observed data distribution with a Dirichlet process mixture of multivariate normals. The model is flexible to capture many distributional features of the observed data, and can be easily modified to reflect different distributions for imputation of the nonrespondents in case of nonignorable missingness. With the development of the NIMC application, the user can easily generate several scenarios to be compared through sensitivity analysis.

Our method is appropriate for sensitivity analysis, and provides a framework to evaluate the impact of extreme scenarios on the results of the imputation. The NIMC application facilitates the process of creating different imputation scenarios, which is an important part of the adaptive design method that we developed. By comparing the scenarios, the user can evaluate the costs and benefits of collecting more data. These benefits can be quantified by the utility measures that we propose, based on propensity scores and marginal statistics of the variables of interest, for different sample sizes.

If the decision is to stop data collection after considering the measures under different scenarios, the nonrespondents have to be imputed for release of a completed data set. In that case, the user might consider using external information or even a different imputation model. For example, in the CMF there are some data available from administrative records about employment, sales and payroll, collected from other sources, such as the Internal Revenue Service (IRS). These administrative records are highly correlated to the actual recorded responses. Thus, they could be modeled jointly with the variables of interest. The missing data can then be imputed conditioned on these other variables, similarly to the item nonresponse imputation that we described in Section 2.2. Here, we assume that given other variables, the missing items are missing at random. A natural extension of this method is

to enable imputation for nonignorable item nonresponse. Another direction for future work is to consider nonresponse in the follow-up sample, which would arise in practice.

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References

Diggle, P. and Kenward, M. G. (1994). Informative drop-out in longitudinal data analysis. *Applied statistics*, 43(1):49–93.

Escobar, M. D. and West, M. (1995). Bayesian density estimation and inference using mixtures. *Journal of the American Statistical Association*, 90(430):577–588.

Ferguson, T. S. (1973). A Bayesian analysis of some nonparametric problems. *The Annals of Statistics*, 1(2):209–230.

Ferguson, T. S. (1983). Bayesian density estimation by mixtures of normal distributions. *Recent advances in statistics*, 24:287–302.

Finamore, J., Reist, B., and Coffey, S. (2013). 2013 National survey of college graduates: A practice-based investigation of adaptive design. In *AAPOR (American Association for Public Opinion Research) 68th Annual Conference*.

Fraley, C. and Raftery, A. E. (2007). Bayesian regularization for normal mixture estimation and model-based clustering. *Journal of Classification*, 24(2):155–181.

Greenlees, J. S., Reece, W. S., and Zieschang, K. D. (1982). Imputation of missing values when the probability of response depends on the variable being imputed. *Journal of the American Statistical Association*, 77(378):pp. 251–261.

Groves, R. (2004). *Survey Errors and Survey Costs*. Wiley Series in Probability and Statistics. Wiley, New York.

Ishwaran, H. and James, L. F. (2001). Gibbs sampling methods for stick-breaking priors. *Journal of the American Statistical Association*, 96(453).

Kim, H. J., Reiter, J. P., Wang, Q., Cox, L. H., and Karr, A. F. (2014). Multiple imputation of missing or faulty values under linear constraints. *Journal of Business & Economic Statistics*, 32(3):375–386.

Miller, P. V. (2013). Adaptive design at the Census Bureau - a new way of doing business. In *AAPOR (American Association for Public Opinion Research) 68th Annual Conference*.

Müller, P. and Mitra, R. (2013). Bayesian nonparametric inference – why and how. *Bayesian Analysis*, 8(2):269–302.

Murphy, S. A. (2003). Optimal dynamic treatment regimes. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 65(2):331–355.

R Core Team (2013). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria.

Rao, R. S., Glickman, M. E., and Glynn, R. J. (2008). Stopping rules for surveys with multiple waves of nonrespondent follow-up. *Statistics in medicine*, 27(12):2196–2213.

RStudio and Inc. (2014). *shiny: Web Application Framework for R*. R package version 0.9.1.

Rubin, D. B. (1976). Inference and missing data. *Biometrika*, 63(3):581–592.

Schouten, B., Bethlehem, J., Beullens, K., Kleven, ., Loosveldt, G., Luiten, A., Rutar, K., Shlomo, N., and Skinner, C. (2012). Evaluating, comparing, monitoring, and improving representativeness of survey response through r-indicators and partial r-indicators. *International Statistical Review*, 80(3):382–399.

Schouten, B., Cobben, F., and Bethlehem, J. (2009). Indicators for the representativeness of survey response. *Survey Methodology*, 35(1):101–113.

Schouten, B., Shlomo, N., and Skinner, C. (2011). Indicators for monitoring and improving representativeness of response. *Journal of Official Statistics*, 27(2):1–24.

Sethuraman, J. (1994). A constructive definition of Dirichlet priors. *Statistica Sinica*, 4:639–650.

Wagner, J. (2008). *Adaptive Survey Design to Reduce Nonresponse Bias*. PhD thesis, University of Michigan.

Wagner, J. and Raghunathan, T. E. (2010). A new stopping rule for surveys. *Statistics in medicine*, 29(9):1014–1024.

West, M., Müller, P., and Escobar, M. D. (1994). Hierarchical priors and mixture models, with applications in regression and density estimation. *Aspects of Uncertainty: A Tribute to D. V. Lindley*, pages 363–386.

Woo, M.-J., Reiter, J. P., and Karr, A. F. (2008). Estimation of propensity scores using generalized additive models. *Statistics in Medicine*, 27(19):3805–3816.

Woo, M.-J., Reiter, J. P., Oganian, A., and Karr, A. F. (2009). Global measures of data utility for microdata masked for disclosure limitation. *Journal of Privacy and Confidentiality*, 1(1):7.